

Project overview

XR piano learning

Unity + Meta Quest 3 + AKAI MPK mini

Learning experience

Falling notes, MIDI input, and immediate performance feedback.

Current stage

A modular functional demo for beginner piano exercises.

Code structure

Rhythm/Core	Lesson models, shared clock, session state, and scoring.
Rhythm/Unity	JSON loading, controller, falling notes, and feedback.
Editor/MIDI + MIDI/ Network	Mac USB input and the Wi-Fi relay.
XR	Hand tracking, gesture controls, and headset presentation.
Resources/Lessons · Scenes · Tests	Lesson content, separate demos, and verification.

Lesson engine

Structure

Lesson · LessonSession · MusicalClock · OnsetEvaluator

Main functions

Start / Pause / Resume / Restart manage the session.
Submit evaluates input; Advance detects missed notes.

Connections

Receives NoteEvent input and produces performance results.
One clock drives both scoring and visualization.

Design

Pure C# logic, independent of Unity and hardware.

Unity controller

Structure

RhythmDemoController + JsonLessonRepository

Main functions

Load reads lesson JSON. The controller initializes modules and handles Start, Pause, Resume, and Restart.

Connections

Queued input is processed before scoring and rendering. Evaluation events trigger visual and text feedback.

Design

Coordinates modules; scoring stays in the lesson engine.

Presentation: notes and feedback

Structure

FallingNotePanel + DemoFeedback

Main functions

Render positions notes using lesson time. ApplyFeedback updates visuals; Complete formats the performance summary.

Connections

Reads lesson data, clock time, and evaluation results. Buttons send commands to the controller.

Design

Displays results without deciding musical correctness.

MIDI input and Wi-Fi relay

Structure

MacMidiInput · MidiByteParser · UdpMidiSender
UdpMidiReceiver · QuestMidiRelay

Main functions

Poll keyboard bytes, normalize NoteEvent data, send packets, and validate received events.

Connections

Keyboard → Mac USB → Wi-Fi → Quest adapter → controller

Status

Relay implemented; complete physical playing test pending.

XR hand tracking and interaction

Structure

QuestHandGestureProvider · PinchGestureState
GestureLessonBinding · QuestLessonControls

Main functions

Read hand joints; detect pinch, release, and tracking loss.
Head-gaze dwell activates fallback buttons.

Connections

Pinch starts a ready lesson. UI controls call the same
controller methods used by desktop controls.

Boundary

Gestures control the lesson; MIDI identifies played notes.
Finger-to-key attribution is not yet implemented.

Content, scenes, and testing

Structure

Resources/Lessons stores JSON; Scenes separates desktop, gesture, simulated Quest, and Wi-Fi relay demos.

Main functions

Scene builders assemble components. Automated tests check timing, scoring, restart, gesture states, and network behavior.

Connections

Each demo uses the same lesson engine with different input and presentation adapters.

Benefit

Test lesson logic independently, then verify hardware on Quest.

Implemented lesson flow

- 1 Load** Read a JSON lesson: C4–G4, five notes at 60 BPM.
- 2 Synchronize** The shared clock drives falling notes and evaluation.
- 3 Evaluate** Check incoming MIDI events for pitch and onset timing.
- 4 Respond** Display feedback and completion summary.
Support pause, resume, and restart.

MIDI and XR integration

Input path

Keyboard → Mac USB → Wi-Fi → Quest lesson

Hand interaction

Quest detects pinch, release, and tracking loss.

Lesson controls

Pinch starts a ready lesson; head-gaze buttons provide fallback.

Current boundary

Pinch detection is implemented. Identifying which finger pressed a physical piano key is future work.

Current testing versions

Milestone1Demo	Desktop lesson; physical MIDI previously verified.
QuestGestureTest	Hand tracking and pinch verified on Quest.
QuestLessonDemo	Quest lesson with simulated notes; user-tested.
QuestMidiRelayDemo	Installed; Wi-Fi acknowledgement verified. Physical playing pending.
Latest checks	26/26 targeted tests passed. Current issue: Unity reports no MIDI device, although macOS detects the MPK mini.

Next steps

1 Stabilize input

Resolve MIDI monitor detection; verify the full keyboard-to-Quest flow and measure latency while playing.

2 Align the experience

Calibrate note lanes to the physical keyboard.
Test chord-to-start and clearer hand feedback.

3 Expand the application

Investigate finger attribution and laptop-free MIDI.
Implement the Figma UI and develop adaptive lessons.